

Slice sampling

The sampling identity, the algorithm, and why it works

Econometrics 32706

The problem

We want draws from a density known only up to a normalizing constant:

$$\pi(x) = \frac{q(x)}{Z}, \quad Z = \int_{\mathbb{R}} q(t) dt.$$

We can evaluate $q(x)$, but calculating Z or sampling directly from π may be difficult.

Key idea: represent π as the marginal distribution of a point chosen uniformly from a region under the graph of q . This representation leads to slice sampling.

Proposition: uniform sampling under a graph

Let $q : \mathbb{R} \rightarrow [0, \infty)$ be measurable and $0 < Z = \int_{\mathbb{R}} q(x) dx < \infty$. Define

$$A = \{(x, h) \in \mathbb{R} \times (0, \infty) : 0 < h < q(x)\}.$$

Sampling proposition

If (X, H) is uniform on A with respect to area, then X has density

$$\pi(x) = \frac{q(x)}{Z}.$$

The height H is auxiliary: discard it after sampling (X, H) , and the remaining X follows the target distribution.

Source: Neal (2003), *Slice sampling*, *Annals of Statistics* 31(3), 705–767, doi:10.1214/aos/1056562461.

Proof of the sampling proposition

Adding vertical strips of height $q(x)$ (Tonelli's theorem) gives

$$|A| = \int_{\mathbb{R}} \int_0^{q(x)} dh \, dx = \int_{\mathbb{R}} q(x) \, dx = Z.$$

Since $|A| = Z$, the uniform density on A is

$$p(x, h) = \frac{1}{|A|} \mathbf{1}\{(x, h) \in A\} = \frac{1}{Z} \mathbf{1}\{0 < h < q(x)\}.$$

Its marginal density for X is π :

$$p_X(x) = \int_0^{q(x)} \frac{1}{Z} \, dh = \frac{q(x)}{Z} = \pi(x).$$

Conclusion 1: sample (X, H) from this joint density and keep only X ; then $X \sim \pi$.

Conclusion 2: alternate draws from $H \mid X$ and $X \mid H$ to preserve this joint density.

Conditional 1: the height given the position

From slide 4,

$$p(x, h) = \frac{1}{Z} \mathbf{1}\{0 < h < q(x)\}, \quad p_X(x) = \frac{q(x)}{Z}.$$

For a fixed x with $q(x) > 0$, divide the joint density by the marginal density of X :

$$p(h | x) = \frac{p(x, h)}{p_X(x)} = \frac{1}{q(x)} \mathbf{1}\{0 < h < q(x)\}.$$

So:

$$H | X = x \sim \text{Unif}(0, q(x)).$$

Conditional 2: the position given the height

By definition, for a fixed height $h > 0$,

$$x \in S_h \iff q(x) > h \iff 0 < h < q(x) \iff (x, h) \in A.$$

Therefore the joint density can also be written as

$$p(x, h) = \frac{1}{Z} \mathbf{1}\{x \in S_h\}.$$

For heights with $|S_h| > 0$, the height marginal is $p_H(h) = |S_h|/Z$. Dividing gives

$$p(x | h) = \frac{p(x, h)}{p_H(h)} = \frac{1}{|S_h|} \mathbf{1}\{x \in S_h\}.$$

So:

$$X | H = h \sim \text{Unif}(S_h).$$

How to use it

The two conditional laws are

$$H \mid X = x \sim \text{Unif}(0, q(x)) \quad \text{and} \quad X \mid H = h \sim \text{Unif}(S_h).$$

Alternate draws from these two conditionals. The resulting Markov chain has stationary joint density

$$p(x, h) = \frac{1}{Z} \mathbf{1}\{(x, h) \in A\}.$$

The ideal slice-sampling update

Starting from the current state value x :

1. Draw $U \sim \text{Unif}(0, 1)$ with realization u , then set $h = uq(x)$.
2. Form the horizontal slice $S_h = \{z : q(z) > h\}$.
3. Draw $X' \mid H = h \sim \text{Unif}(S_h)$.
4. Use the realized value x' as the next state and repeat.

A simple numerical example

Let $q(x) = e^{-x^2/2}$, the standard normal kernel, and suppose the current value is $x = 1$. For a realization $u = \frac{1}{2}$ of $U \sim \text{Unif}(0, 1)$, the height is

$$h = u q(1) = \frac{1}{2} e^{-1/2}.$$

The slice condition is

$$e^{-z^2/2} > \frac{1}{2} e^{-1/2} \quad \Longleftrightarrow \quad |z| < \sqrt{1 + 2 \log 2} \approx 1.54.$$

Thus $S_h \approx (-1.54, 1.54)$, and $X' \mid H = h \sim \text{Unif}(S_h)$ has a realized value x' in this interval.

The example illustrates one possible height; the next update draws a new one.

Why the update targets π

Start with $X \sim \pi$. Given its current value x , draw $H \mid X = x$; then (X, H) has joint density $p(x, h)$. Next draw $X' \mid H = h$. This leaves the joint law unchanged, so discarding H gives $X' \sim \pi$.

This is a Gibbs-sampling argument. It proves that π is stationary for the update; successive states need not be independent.

When the slice is hard to sample from

The ideal update assumes we can draw uniformly from S_h . For a complicated density, its endpoints may be unknown and S_h may have several pieces.

A practical one-dimensional slice sampler constructs a bracket around the current point, tests candidate positions, and shrinks the bracket after rejections. The bracket rules must preserve the correct conditional distribution on the slice.

The height step remains $h = u q(x)$. If we work with log densities, the equivalent threshold is

$$\log h = \log q(x) + \log u.$$

This form avoids evaluating a tiny density value directly.

What to remember

1. Uniform sampling under an unnormalized density curve gives the desired marginal distribution.
2. A random height selects a horizontal slice; the position update moves within that slice.
3. Alternating these conditional updates preserves the joint law and therefore the target marginal.

Reference: Radford M. Neal (2003), *Slice sampling*, *Annals of Statistics* 31(3), 705–767, doi:10.1214/aos/1056562461.

ESS uses the same slice idea

Let $\phi_K(f) = \mathcal{N}(f; 0, K)$. The target is

$$\pi(f) \propto \phi_K(f)L(f).$$

Introduce a height *under the likelihood*:

$$p(f, h) \propto \phi_K(f) \mathbf{1}\{0 < h < L(f)\}.$$

Ordinary slice sampling recovers its target by integrating out height:

$$p_X(x) = \int_0^{q(x)} \frac{1}{Z} dh = \frac{q(x)}{Z} = \pi(x).$$

ESS uses the same step, with the prior factor inside the integral:

$$p_F(f) \propto \int_0^{L(f)} \phi_K(f) dh = \phi_K(f)L(f) \propto \pi(f).$$

The two conditional distributions follow by dividing this joint density by its marginals.

The two conditional updates for ESS

Let $Z_* = \int \phi_K(z) L(z) dz$, the normalizing constant. The joint density is

$$p(f, h) = \frac{1}{Z_*} \phi_K(f) \mathbf{1}\{0 < h < L(f)\}.$$

The marginal density of F is $p_F(f) = \phi_K(f) L(f) / Z_*$, so

$$p(h | f) = \frac{p(f, h)}{p_F(f)} = \frac{1}{L(f)} \mathbf{1}\{0 < h < L(f)\}, \quad H | F = f \sim \text{Unif}(0, L(f)).$$

At a fixed height, define

$$C_h = \int \phi_K(z) \mathbf{1}\{L(z) > h\} dz.$$

Integrating over this slice gives $p_H(h) = C_h / Z_*$. Dividing by this marginal gives

$$p(f | h) = \frac{\phi_K(f) \mathbf{1}\{L(f) > h\}}{C_h} = \pi_h(f), \quad F | H = h \sim \pi_h.$$

ESS searches that slice along an ellipse

From the current value f , draw $V \sim \mathcal{N}(0, K)$ and $U \sim \text{Unif}(0, 1)$. Let v and u be their realized values, set $h = uL(f)$, and consider

$$f(\theta) = f \cos \theta + v \sin \theta, \quad S_h^\theta = \{\theta : L(f(\theta)) > h\}.$$

A rotation of the *pair* is

$$(f_\theta, v_\theta) = (f \cos \theta + v \sin \theta, -f \sin \theta + v \cos \theta).$$

ESS accepts only angles in S_h^θ . It shrinks a randomly placed angular bracket after rejected candidates; the next two slides show why this gives a valid update for $p(f \mid h)$.

Since $f(0) = f$ and $h < L(f)$, angle 0 is always eligible. The bracket keeps it inside while searching.

The conditional distribution at fixed h

Let ϕ_K be the $\mathcal{N}(0, K)$ density and define

$$C_h = \int \phi_K(z) \mathbf{1}\{L(z) > h\} dz.$$

Add an independent Gaussian draw V to the conditional target:

$$\rho_h(f, v) = \frac{1}{C_h} \phi_K(f) \phi_K(v) \mathbf{1}\{L(f) > h\}.$$

Its marginal density for F is exactly the distribution we need:

$$\int \rho_h(f, v) dv = \frac{\phi_K(f) \mathbf{1}\{L(f) > h\}}{C_h} = p(f | h).$$

Also, $\rho_h(v | f) = \phi_K(v)$. Thus drawing $V \sim \mathcal{N}(0, K)$ is an exact conditional draw for this augmented target.

Why the ESS angle search preserves it

Rotating (f, v) by θ to (f', v') has unit Jacobian and preserves the Gaussian product:

$$\phi_K(f')\phi_K(v') = \phi_K(f)\phi_K(v).$$

ESS accepts only $L(f') > h$; both endpoints of an accepted move therefore have the same density under ρ_h .

The reverse rotation uses $-\theta$. The randomly placed initial bracket and the shrinkage rule give matching probabilities to the forward and reverse angle paths (Murray, Adams and MacKay, 2010, Section 2.3). Hence the move satisfies detailed balance for ρ_h .

Consequently,

$$F \mid H = h \sim \pi_h \quad \implies \quad F'_{\text{ESS}} \mid H = h \sim \pi_h.$$

This is a valid conditional *Markov update*, not an independent draw from π_h .

Ordinary slice sampling versus ESS

	Ordinary slice sampling	Elliptical slice sampling
Target	$\pi(x) \propto q(x)$	$\pi(f) \propto \phi_K(f)L(f)$
Height	$H \mid X = x \sim \text{Unif}(0, q(x))$	$H \mid F = f \sim \text{Unif}(0, L(f))$
Given h	Uniform on $\{x : q(x) > h\}$	Gaussian prior restricted to $\{f : L(f) > h\}$
Update	Ideal: uniform conditional draw	Reversible Markov update on the restricted prior

Shared principle: augment with a height, then update while preserving the conditional law on that slice.

ESS source: Murray, Adams and MacKay (2010), *Elliptical slice sampling*, AISTATS, PMLR 9, 541–548, proceedings.mlr.press/v9/murray10a.html.